The GPS pages

GPS interference as the aircraft report it — Quality, Episodes and Aircraft

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In short. Every aircraft with ADS-B tells in its own messages how good its GPS position is. MODE-S Wind listens to these messages from all the aircraft its receivers hear, notices when the quality drops or the position disappears, and keeps a record of it. Three pages show the record:

- **GPS Quality** (`/gps`): the degradation hour by hour and day by day, by altitude band, the aircraft degraded right now, and the severity and spoofing indicators (Section 2).
- **GPS Episodes** (`/gps-episodes`): every continuous period of degradation of one aircraft — where it began, where the aircraft recovered or was lost, at what altitude and for how long — on a map and in a list (Section 3).
- **GPS Aircraft** (`/gps-aircraft`): the good and bad days of *one* aircraft, a callsign or a whole type (Section 4), and the **overview of all traffic**: by day and week, by time of day, for arrivals and departures, by direction and altitude, by kind of degradation, by type, airline and aircraft (Section 5).

Section 1 explains what is measured, Section 6 how to read it sensibly and Section 7 what the first month of data shows.

Full view only, research only. The GPS pages are shown only to users with the **Full view** right; for others the GPS link of the top bar is not there. They describe what the aircraft broadcast, as heard by MODE-S Wind's receivers — they do not name a cause and are not an official source. Military, government and police aircraft are never shown or counted. **All figures show real data**: the stored data of 9 October – 7 November 2026 drawn in a test with the same page code as the live site, as it was on 7 November 2026 at 12:31 UTC (light theme, the top bar left out).

1 What is measured

1.1 The aircraft's own quality figures

An ADS-B transponder broadcasts the aircraft's GPS position about twice a second, and with it two figures that the avionics compute themselves:

- **NACp** (navigation accuracy category, position) — how accurate the position is. 9–11 is normal for GPS (better than 30 m); 7–8 is still good (better than 0.1 NM); **6 or less** means worse than about 0.1 NM, and **0** that the accuracy is unknown.
- **NIC** (navigation integrity category) — how sure the system is that the position is within a radius. 7 or more is normal; **6 or less** means a containment radius of 0.3 NM or more; **0** no integrity at all. NIC comes with every position message, so it reacts within a second.

The full scales are in Appendix A.

1.2 Five degradation signals

Every five seconds MODE-S Wind looks at every aircraft heard by its own receivers above 1 000 ft and raises a signal when one of these holds:

Signal	Condition	What it means
NACp	NACp 6 or less (fresh, last 30 s)	the GPS reports a degraded accuracy
NIC	NIC 6 or less (fresh)	the GPS reports a degraded integrity — often the onset or the fringe of jamming
Freeze	the aircraft keeps sending positions, but the same one in three looks in a row while moving	the GPS output is stuck
Gap	the aircraft transmits ADS-B but has no position from any source for 45 s	GPS lost, and no multilateration (MLAT) either
ADS-B (loss)	no own ADS-B position for 45 s while MLAT still tracks it	GPS lost — the most common signal near EFHK

A degraded five-second look at one aircraft is an **event**; an aircraft degraded for a minute gives about 12 events. The look is **severe** when the position is unusable: NACp 0, NIC 0 or no own position at all.

1.3 Episodes

An **episode** is one continuous period of degradation of one aircraft within **100 NM of EFHK** and above 1 000 ft (short clean breaks under two minutes are joined). It ends as

- **recovered** — two minutes without any signal; the first clean position is the *recovery point*;
- **lost** — the aircraft disappeared while degraded (*left coverage* when that happened at the coverage edge, 130 NM or more);
- **below min. alt.** — it descended below 1 000 ft while still degraded, typically on final.

Each episode keeps its start, end and recovery positions, altitudes and distances, the signals seen, the lowest NACp and NIC and the spoofing indicators.

1.4 Spoofing indicators

Jamming takes the GPS position away; *spoofing* replaces it with a false one, and a spoofed receiver may report perfectly good NACp and NIC. Four checks are recorded beside the signals (they do not count as degradation): *ADS-B / MLAT position mismatch* (the aircraft's position and the multilateration position disagree by more than 2 NM twice in a row), *GNSS − baro at its limit* and *GNSS − baro jump* (the difference between the GPS and the pressure altitude, which normally changes slowly, is at its largest value or jumps by 500 ft or more), and *position jump* (the position steps further than the speed allows, confirmed by the next one). Agreeing indicators from **two or more independent checks** (`2+`) are much stronger evidence than one alone; a single indicator during jamming is more often the receiver re-acquiring the satellites.

1.5 Flights and passes

For the GPS Aircraft page a **flight** is one *pass* of an aircraft through the receivers' coverage. The passes come from the dense altitude series that MODE-S Wind stores for each aircraft visit — but that series needs the aircraft's GPS height, which disappears while GPS is lost. So the **episodes are passes too**: a flight is the aircraft's stored passes and episodes of one callsign less than 20 minutes apart. A flight is **degraded** when it has at least one episode. From 8 November 2026 each pass also keeps where it entered and left the coverage (bearing and distance from EFHK, altitude), kept for 400 days.

2 The GPS Quality page



Figure 1. GPS Quality: the summary bar at the top, the events of the last 24 hours (left), the FL-band heatmap of 14 days (left bottom), the live table, the FL-band analysis and the severity and spoofing panel (right). 7 November 2026, 12:31 UTC.

Summary bar. The Quality | Episodes | Aircraft switch, then the events and the aircraft affected in the last 24 hours, Severe 24h (aircraft-hours with an unusable position), the peak hour, the aircraft degraded now and the **zone**: All (the whole coverage), 50 nm or 20 nm around the airport.

Events chart. One bar per hour (1d–1w) or per day (2w–6m), split by signal: NACp orange, NIC rose, Freeze blue, Gap violet, ADS-B teal. The grey line is the traffic (aircraft per hour) and the dashed gold line *events per aircraft* — the degradation index that compares days with different traffic. 10 min switches to the degraded aircraft per ten minutes, *severe* red and *mild* amber, with the share of the traffic at FL050–250 degraded. Dashed amber markers show a change of the counting method; values on the two sides are not directly comparable.

FL-band heatmap. Nine altitude bands from FL010 to above FL300, one column per day (14d / 1m): Events counts the events (each row on its own scale), % aircraft the share of the aircraft-hours in that band with a degradation. Clicking a cell opens GPS Episodes for that day and band.

Right column. The *live table* lists the aircraft degraded now (a red callsign: severe). *FL band analysis* gives the share of

the events by band, the worst day and the mix of signals. *Severity & Spoofing* (1d / 7d / 1m) gives the degraded and severe aircraft-hours, each spoofing indicator with the aircraft, how many of them reported normal GPS quality at the time (the strong spoofing case, in yellow) and how often the check could be made, the 2+ count, and the list of flagged aircraft.

3 The GPS Episodes page

Control bar. Window 3 h … 1 w (episodes that started in it), or one UTC day by date, Live back to the window; filters Min duration (default 1 min — shorter ones are usually a single NIC or NACp blip), Signal, End, Altitude and Spoofing.

Map. The airport with rings at 20, 50 and 100 NM and the coverage edge at 130 NM. Each episode is a line from where it started (hollow circle) to the last degraded position (filled circle), coloured by its end: recovered green (a dashed line to the recovery point), lost red, left coverage grey, below min. alt. gold.

List and chart. Newest first: time, callsign, type, duration, the signal chips (NACp, NIC, FRZ, GAP, LOSS, and SPF / SPF 2+ for spoofing indicators), the flight level and the distance at start → end (⋆ = it started at the 100 NM edge, so the aircraft was most likely already degraded when it came in) and the end. Clicking a row selects the episode on the map and opens its details. *Altitude vs distance* plots each episode's start (▲), recovery point (●) and lost position (×) as altitude against the distance from the airport.

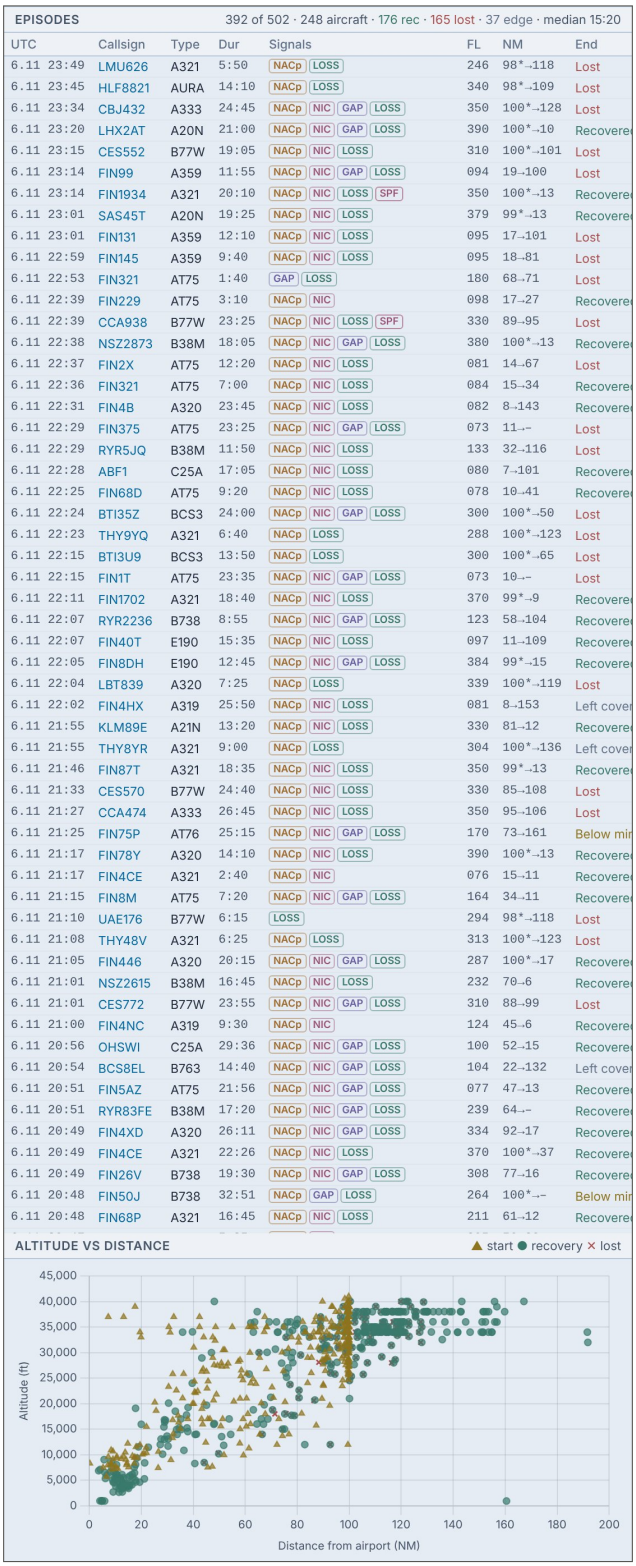


Figure 2. GPS Episodes, 6 November 2026: the list (392 of 502 episodes of at least one minute, 248 aircraft) and *altitude vs distance*. Most episodes start at about 100 NM — the aircraft came in already degraded — and the recovery points low near the airport are mostly arriving aircraft. The map beside the list is not shown here.

4 GPS Aircraft: one aircraft, a callsign or a type

Type in the search box an **ICAO24 address** (`4601FA`), a **registration** (`OH-ATM`), a **callsign** (every aircraft that flew it) or an **ICAO type** (`AT75` — the whole fleet seen) and press Show. Period 7 / 30 / 60 / 90 days or All (the episodes are kept from 6 October 2026). The address keeps the choice (`/gps-aircraft?q=AT75&days=30`), so it can be bookmarked or sent. A callsign in the GPS Episodes list opens its aircraft here.

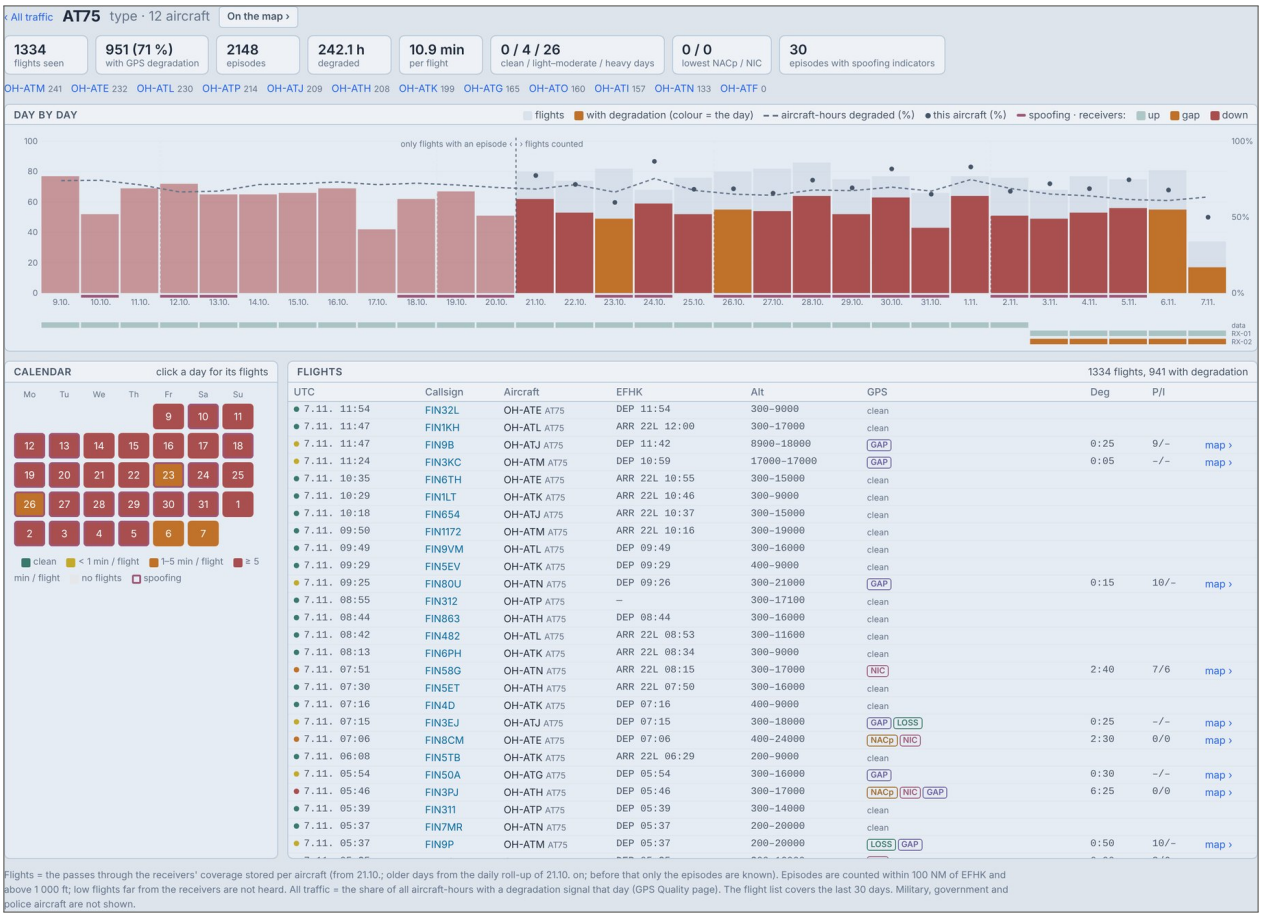


Figure 3. GPS Aircraft for the type AT75 (ATR 72-500), 30 days: 12 aircraft, 1 334 flights, 71 % of them with GPS degradation. The days left of the dashed line are before the flight counts (only the flights with an episode are known); the strip under the chart is the receivers' coverage.

- **Key figures** — the flights seen, the flights with degradation and their share, the episodes, the degraded time and minutes per flight, the clean / light–moderate / heavy days, the lowest NACp / NIC and the episodes with a spoofing indicator. On the map › opens the same episodes on GPS Episodes; ‹ All traffic returns to the overview.
- **Day by day** — per UTC day the flights seen (grey) and in front of them the flights with degradation, coloured by the day: **clean** green, under 1 degraded minute per flight yellow, 1–5 minutes orange, **5 minutes or more** red; a pink mark for spoofing indicators. The dots are this aircraft's share of flights degraded, the dashed line the day's general level — the share of all aircraft-hours with a degradation signal. A bad day for everyone points to jamming; a bad day for this aircraft only to its equipment or its route.
- **Receiver coverage** — the thin rows under the chart: each receiver's minutes with messages per day (RX-01, RX-02 …) and, before the per-receiver record began, the hours with any data. Pale: up (23 hours or more); orange: a gap; red: down. On a day with a gap fewer flights are heard, especially low ones far from the airport.
- **Calendar** — the same colours by week; clicking a day lists its flights.
- **Flights** — the flights of the last 30 days, newest first: time, callsign, the landing (`ARR 22L 10:46`) or departure at EFHK, the altitude range, the signals, the degraded time, the lowest NACp / NIC and map ›.

5 GPS Aircraft: the overview of all traffic

With the search box empty the page shows **all traffic**. Three choices in the control bar filter the whole overview (and stay in the address, `/gps-aircraft?days=30&h=0-6&al=FIN`):

- Period — as above.
- Hours (UTC) — All day, 00–06, 06–12, 12–18, 18–24 or From – to…: only the flights and episodes that *started* in those hours (also across midnight).
- Airline — only the flights of one airline, by the ICAO code at the start of the callsign (`FIN`, `SAS` …, with the number of flights), or *other* (callsigns without one). With one airline, the Types table compares the airline's own types on the same routes. Clicking an airline in the Airlines table chooses it; ‹ All airlines returns.



Figure 4. The control bar, the key figures and the days of all traffic, 30 days.

Key figures — flights seen, flights with GPS degradation, episodes, degraded hours and minutes per flight, **today so far** against the median of the last seven days, the best and the worst day (of the days with at least 20 flights and no receiver gap), the mean share of aircraft-hours degraded (the GPS Quality measure) and the episodes with spoofing indicators.

All traffic, day by day — as an aircraft's chart; the degraded flights are coloured by the share of the day's flights (light → dark red), with the receivers' coverage below.

Week by week — the share of the flights degraded per ISO week (Monday–Sunday) with the minutes per flight; ⋆ marks a week only partly in the period.

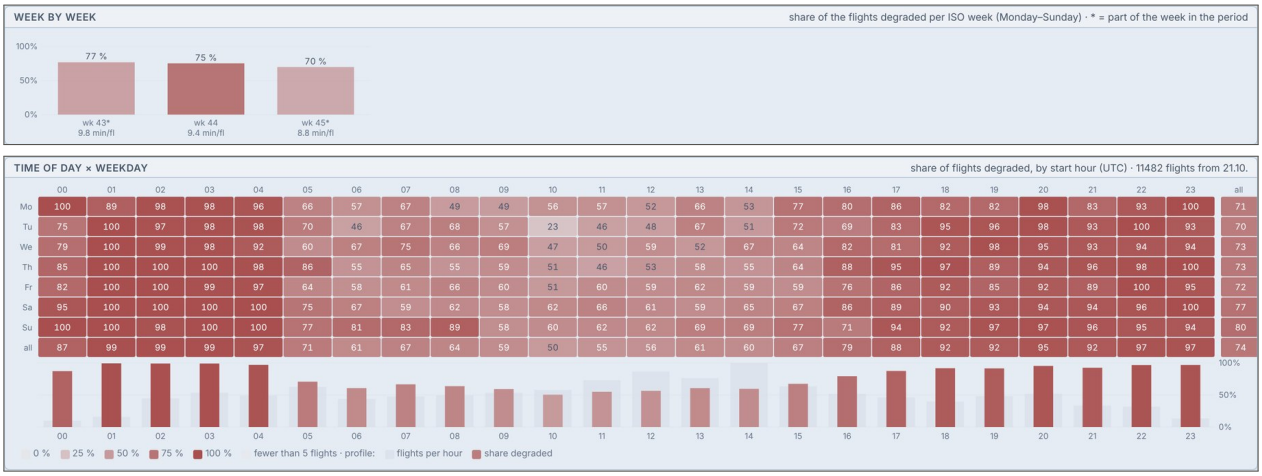


Figure 5. Week by week, and the share of the flights degraded by start hour (UTC) and weekday, with the totals and the hour profile.

Time of day × weekday — the share of the flights degraded by the hour (UTC) the flight started and its weekday; the totals per weekday on the right (*all*) and per hour in the bottom row, and an hour profile below (the flights per hour in grey, the share degraded in colour). A cell with fewer than five flights is left uncoloured. Hover a cell for its numbers.

Arrivals and departures at EFHK — an *arrival* is a flight that ends within 10 NM of EFHK at or below 4 000 ft, a *departure* one that starts so; the rest are overflights (also other airports' traffic). For the arrivals: how many were degraded on the way in, **where GPS came back** (the median distance and altitude of the recovery point of the last episode), how many got it back only within 30 NM and how many were **still degraded below 1 000 ft**. The bars give each day's arrivals as *clean*, *GPS back beyond 30 NM*, *GPS back within 30 NM* and *degraded below 1 000 ft*. The histograms show where GPS came back on arrival and where it was lost on departure, in 10 NM steps.

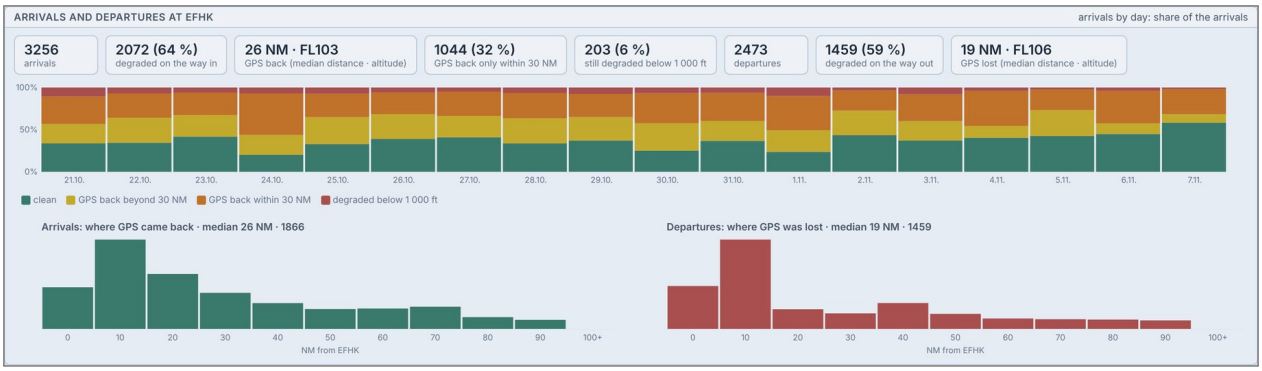


Figure 6. Arrivals and departures, 21 October – 7 November 2026: 3 256 arrivals, 64 % degraded on the way in; GPS came back at a median of 26 NM and FL103; 203 arrivals (6 %) were still degraded below 1 000 ft.

By direction and altitude — three compass roses, each sector coloured by the share of its flights degraded, with the share and the number of flights: *arrivals* by where they entered the coverage, *departures* by where they left it, *overflights* by where they entered. Hover an arrivals sector for its median recovery distance and how many were still degraded below 1 000 ft. Below, the share degraded by the *highest altitude of the pass* and where the episodes started.

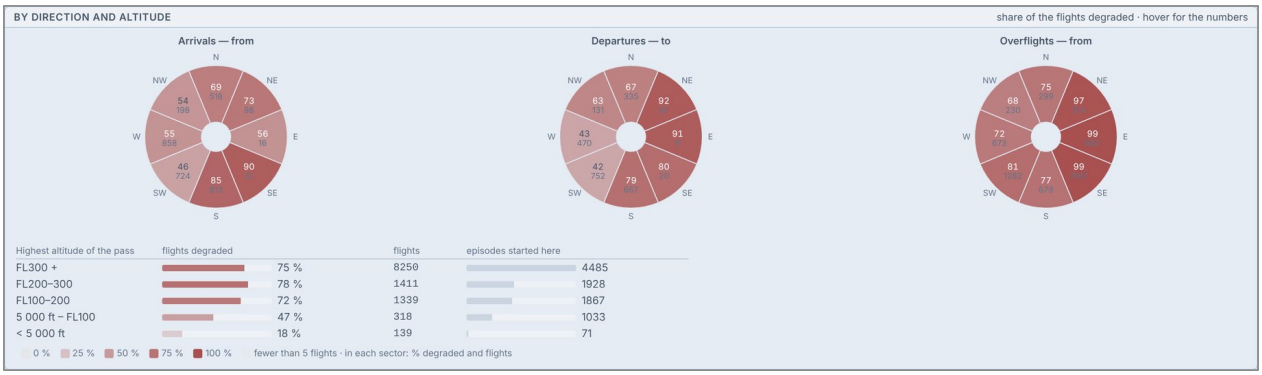


Figure 7. By direction and altitude. Arrivals from the south were degraded most (85 %), from the south-west least (46 %); overflights from the east and south-east almost always (99 %).

Kind of degradation — the episodes by day or by start hour (By day / By hour): *position lost* (the aircraft's own ADS-B position stopped — the ADS-B loss signal), *accuracy reduced* (NACp or NIC only) and *gaps / frozen position only*; a pink mark for the episodes with spoofing indicators, and the indicators by kind.

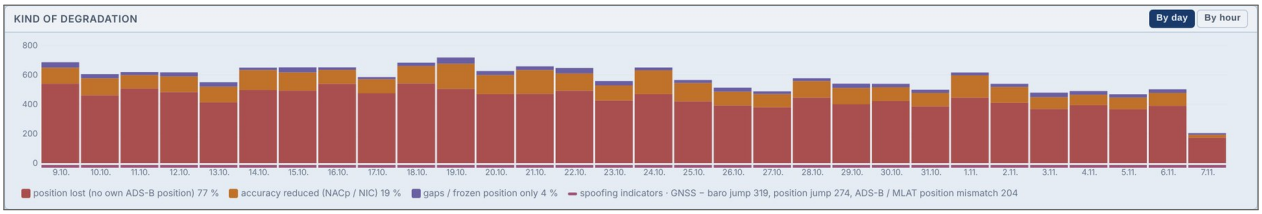
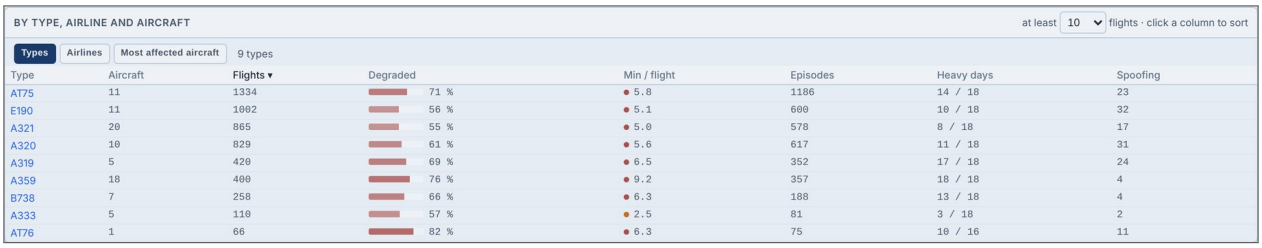


Figure 8. Kind of degradation by day: in about three episodes out of four the aircraft lost its own position.

By type, airline and aircraft — three tabs: Types (the ICAO type), Airlines (the callsign's ICAO code) and Most affected aircraft (the 40 with the highest share degraded). Columns: aircraft, flights, **degraded** (share of the flights), minutes per flight, episodes, **heavy days** (5 or more degraded minutes per flight) of the days flown, spoofing. Only those with at least 10 flights (5 … 100); click a column to sort, a type or an aircraft for its history.



Type	Aircraft	Flights ▾	Degraded	Min / flight	Episodes	Heavy days	Spoofing
AT75	11	1334	71 %	5.8	1186	14 / 18	23
E190	11	1002	56 %	5.1	600	10 / 18	32
A321	20	865	55 %	5.0	578	8 / 18	17
A320	10	829	61 %	5.6	617	11 / 18	31
A319	5	420	69 %	6.5	352	17 / 18	24
A359	18	400	76 %	9.2	357	18 / 18	4
B738	7	258	66 %	6.3	188	13 / 18	4
A333	5	110	57 %	2.5	81	3 / 18	2
AT76	1	66	82 %	6.3	75	10 / 16	11

Figure 9. Types of one airline (Airline FIN), 30 days: the AT75 had GPS degradation on 71 % of its flights, the E190 on 56 % and the A321 on 55 %.

6 How to read it sensibly

- **Flights or aircraft-hours.** The GPS Quality page counts aircraft-hours with a degradation signal; the Aircraft page counts flights with an episode. A long overflight is one flight but several aircraft-hours, so the two shares differ a little (on 21 October – 7 November 74 % of the flights against 67 % of the aircraft-hours).
- **Who is in the air matters.** The share of a time of day or a direction is the share of *its* traffic. At night most traffic is long overflights from the east and south-east, which are degraded almost always, so the night looks worse than the day partly because the traffic is different. Compare like with like: one airline, one type, arrivals only.
- **The monitored area.** Episodes are counted within 100 NM of EFHK above 1 000 ft. An aircraft degraded before it comes in starts its episode at the 100 NM edge ($\star$ in the list), and one that leaves degraded ends as *left coverage*.
- **What the receivers hear.** Low aircraft far from the receivers are not heard, so the low altitude band is mostly the arrivals and departures near EFHK. A receiver gap (the coverage strip) removes flights, mostly far and low ones; such days are left out of the best and worst day. As more receivers come, more low flights at the edges are heard.
- **No cause is named.** Jamming, spoofing, a fault in one aircraft and a real satellite problem can look alike. A bad day for every aircraft points to interference; a bad day for one aircraft only to its equipment or its route. A spoofing indicator alone is weak evidence; `2+` is stronger.
- **Small numbers.** A sector or a type with a handful of flights can show 0 % or 100 % by chance; look at the number of flights beside the share.

7 First results (9 October – 7 November 2026)

From the stored data as on 7 November 2026, 12:31 UTC (the flight counts from 21 October):

- **GPS degradation is the rule, not the exception**: 74 % of the 11 509 flights seen had at least one episode, on average 9.3 degraded minutes per flight; even the best day had 68 %. In about three episodes out of four the aircraft lost its own position altogether.
- **Time of day**: by start hour the share was 97–99 % at 01–04 UTC and 88–97 % at 17–23 UTC, lowest around 10 UTC (50 %) — with the caution above about the night's traffic.
- **Direction**: overflights from the east and south-east 99 %, from the north-east 97 %; arrivals from the south 85 % and from the south-east 90 % (31 flights), from the south-west 46 % and the west 55 %.
- **On the way in**: 64 % of the 3 256 arrivals were degraded; GPS came back at a median of 26 NM and FL103 — for arrivals from the south at 14 NM, from the west at 57 NM. 203 arrivals (6 %) were still degraded below 1 000 ft, 154 of them from the south (19 % of the arrivals from the south).
- **On the way out**: 59 % of the 2 473 departures lost GPS, at a median of 19 NM and FL106.
- **Types**: within one airline (FIN) the AT75 was degraded on 71 % of its flights, the A359 on 76 %, the A320 on 61 %, the E190 on 56 % and the A321 on 55 %.

A NACp and NIC scales

NACp	horizontal accuracy	NIC	containment radius
0	unknown	0	unknown
1 / 2 / 3	< 10 / 4 / 2 NM	1 / 2 / 3	< 20 / 8 / 4 NM
4 / 5 / 6	< 1 / 0.5 / 0.3 NM	4 / 5 / 6	< 2 / 1 / 0.6 NM
7 / 8	< 0.1 / 0.05 NM	7 / 8	< 0.2 / 0.1 NM
9 / 10 / 11	< 30 / 10 / 3 m	9 / 10 / 11	< 75 / 25 / 7.5 m

NACp 6 or less and NIC 6 or less raise a signal.

B Version history

1.0 (8 Nov 2026) First version: the Quality, Episodes and Aircraft pages, the overview of all traffic.